TEXORA · LECTURE SHEET

Process Control in Rotor

Rotor spinning concentrates its control problems inside the rotor itself: deposits that build in the groove, twist efficiency below the calculated value, and the count and strength norms that differ from ring yarn. This sheet covers the rotor-specific measurements and cleaning disciplines.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 10 · SPECIALIZATION — YARN MANUFACTURING

1. The rotor spinning process in control terms

Drawframe sliver is combed open to single fibres by the opening roller, transported by air into the rotor, and deposited as a fibre ring in the rotor groove. The yarn end, fed back through the navel, collects the ring as the rotor turns — twist is inserted between navel and groove. Every element of that path is a control point: opening roller condition governs fibre separation and nep creation; the rotor groove governs deposits; the navel governs twist efficiency and yarn surface.

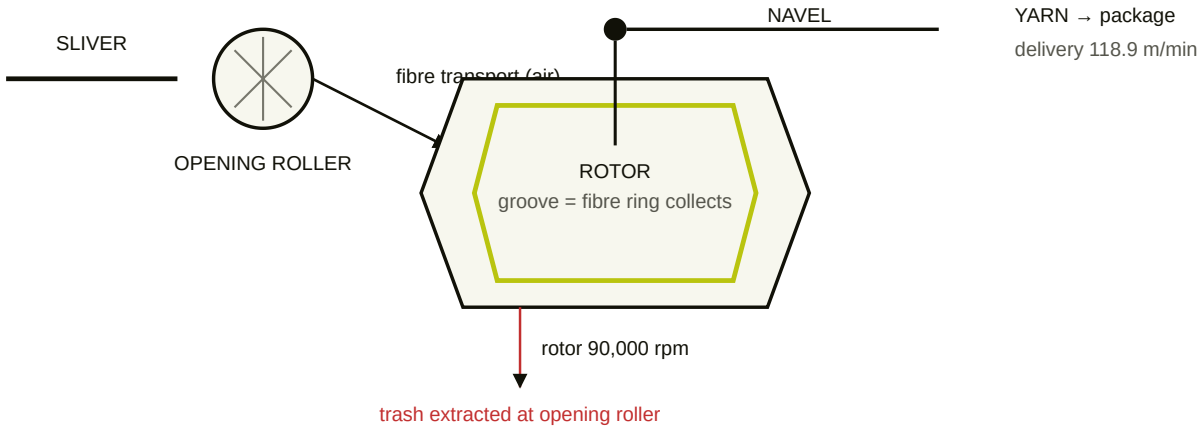


Figure 1 — The rotor spinning path: sliver, opening roller, air transport into the rotor groove, yarn withdrawal through the navel. Deposits collect in the groove; twist is generated between groove and navel.

2. Twist efficiency

Calculated twist follows the YM-3207 relation (rotor rpm ÷ delivery), but the yarn as spun carries less: slip at the navel and the geometry of the false-twist zone mean measured twist falls short. The ratio is the twist efficiency, and it is a monitored quantity because strength follows actual twist, not calculated twist.

WORKED EXAMPLE — TWIST EFFICIENCY

Rotor speed 90,000 rpm; delivery 118.9 m/min (YM-3207 Topic 5).

Calculated twist = 90,000 / 118.9 = **757 turns/m**.

Untwist–retwist testing of the yarn measures **721 turns/m**.

Twist efficiency = 721 / 757 × 100 = **95.2%**.

A fall in twist efficiency at constant settings points to navel wear or groove deposits changing the yarn's path; the corrective is cleaning or navel replacement, not a twist change wheel — the wheel would raise calculated twist while the loss mechanism stays.

3. Rotor deposits — the signature rotor problem

Dust, wax, finish and trash particles carried into the rotor accumulate in the groove. The growing deposit ring disturbs fibre collection, and the yarn shows it as rising CV%, periodic faults, falling strength and increased ends down — often gradually over days, so trend records catch what single tests miss. Control is a cleaning discipline: rotors are cleaned at fixed running-hour intervals (and whenever a head's quality data drifts), with the interval set from the material's dust load — recycled and waste-rich blends shorten it drastically. Deposit mass per rotor at cleaning is itself recorded in some mills as a feed-quality measure.

4. Quality norms for rotor yarn

PROPERTY	ROTOR EXPECTATION (VS RING YARN OF SAME COUNT)	CONTROL NOTE
Count CV%	Comparable or slightly better	Draft is electronic/mechanical at one point; drift usually feed sliver
Strength / CSP	Typically 10–20% lower	Judge against rotor norms, never ring norms; wrapper fibres contribute little strength
Evenness U%	Often better (fewer long-term faults)	Rising CV% trend = deposits until proved otherwise
Neps and trash	Depends on opening roller and feed cleanliness	Trash in feed sliver loads the rotor; opening roller clothing life is a control item
Ends down	Low when clean; climbs with deposits	Head-wise records localise dirty rotors

5. Feed control

Rotor spinning has no intermediate process to forgive feed faults: the drawframe sliver is the direct input, so sliver weight control (Topic 5) is rotor count control, and sliver trash content becomes rotor deposit rate. Mills running rotor frames therefore put their tightest drawframe discipline in front of them and schedule opening-roller clothing changes by throughput, not by calendar.

Glossary

Rotor groove — the V- or T-shaped collecting channel inside the rotor where fibres form a ring. **Navel** — the ceramic guide at the rotor centre through which yarn is withdrawn; its surface generates part of the twist effect. **Twist efficiency** — measured twist as % of calculated (rotor speed/delivery) twist. **Opening roller** — the clothed roller combing sliver into single fibres before transport. **Wrapper fibres** — fibres wrapped around the rotor yarn body, characteristic of its structure; they add bulk but little strength. **Deposit** — accumulated dust, wax and trash compacted in the rotor groove. **Seed coat fragments** — brittle seed-cover particles that survive to the rotor and disturb spinning.

Key points

• Twist efficiency = measured/calculated twist = 95.2% in the example; strength follows the measured value.
• Falling twist efficiency at constant settings means navel wear or deposits — clean or replace, don't re-gear.
• Rotor deposits are the signature fault: rising CV%, periodic faults, ends down — controlled by fixed-interval rotor cleaning.
• Rotor yarn is judged on rotor norms: CSP typically 10–20% below ring yarn of the same count.
• Feed sliver quality is rotor quality: weight control and trash content act directly, with no process between.

Practice

1. Rotor speed is 75,000 rpm and delivery is 105 m/min; the yarn tests 690 turns/m. Calculate the calculated twist and the twist efficiency.
2. A rotor department's yarn CV% climbs slowly over two weeks on all heads, while count stays on target. Give the most likely cause and the control action, and say why a twist increase would be wrong.
3. Why are rotor cleaning intervals shortened when spinning recycled-fibre blends?

Answer key

1. Calculated twist = 75,000/105 = **714 turns/m**. Efficiency = 690/714 × 100 = **96.6%**.

2. Progressive rotor-groove deposit build-up: it disturbs fibre collection (evenness) before it moves the mean count. Action: bring forward the rotor cleaning cycle and verify CV% recovery head group by head group. A twist increase treats strength, not evenness, and would raise neither the deposit nor the CV% problem — it only costs delivery speed.
3. Recycled blends carry more dust, short fibre and finish residues, so deposits accumulate faster in the groove; the cleaning interval must follow the deposit rate of the material actually running, not a standard calendar.
